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Efficient Spatial Channel Estimation in Extremely Large Antenna Array Communication Systems: A Subspace Approximated Matrix Completion Approach

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ABSTRACT Matrix completion techniques are widely employed to estimate the channel matrix from partially observed channel measurements. However, its computational complexity is cubic in the number of antennas, which is non-scalable for extremely large-scale antenna array (ELAA) communication systems. To address this issue, in this paper the ELAA channel matrix completion is reformulated as a proximal gradient descent (PGD) problem, where the subgradient of the nuclear norm is computed by singular value thresholding (SVT) operator and the proximal gradient of the L_1 norm is derived using a soft-thresholding operator. To mitigate the computational overhead caused by subspace orthogonalization in the SVT operation, a novel subspace-approximated (SA)-SVT-PGD algorithm is designed. This algorithm exploits the subspace similarity of the channel's Gram matrix in consecutive PGD iterations and enables concurrent subspace orthogonalization during PGD updates. By eliminating the specific nested loop for the subspace orthogonalization, the computational complexity of the SA-SVT-PGD algorithm is proportional to the product between the number of antennas and the square of rank of the channel matrix. Simulation results demonstrate that the SA-SVT-PGD algorithm can reduce the convergence time by 71.7% compared with the traditional SVT-PGD algorithm.

INDEX TERMS Channel reconstruction, matrix completion, proximal gradient descent, nuclear and L_1 norm optimization, extremely large antenna array.

I. INTRODUCTION

A. MOTIVATIONS

ACCURATE channel estimation is fundamental for enabling precise beamforming, sum-rate maximization, interference management, and effective signal detection in multi-antenna communication systems, which has attracted widespread attention in academia and industry [1], [2], [3], [4], [5], [6], [7]. However, with the deployment of extremely large antenna arrays (ELAA) in fifth-generation advanced (5G-A) base stations that configure 1024 antenna elements to support over one hundred data streams [8], obtaining the complete ELAA channel matrix becomes challenging due to the limited time-frequency resources allocated for reference signals [9].

Matrix completion techniques provide an effective solution for reconstructing the complete channel matrix from partial measurements [10], [11], [12]. By leveraging the inherent low-rank structure and sparsity for the channel matrix, channel matrix completion is typically formulated as a low-rank matrix completion task, whose optimal solution can be obtained using first-order optimization methods such as the alternating direction method of multipliers

(ADMM) [13]. However, traditional low-rank matrix completion techniques often rely on repeated executions of singular value decomposition (SVD), which exhibit cubic computational complexity with respect to the number of antennas [14]. This substantial computational burden and associated latency significantly limit the practicality of deploying traditional matrix completion methods in ELAA communication systems. Consequently, there is a pressing need to develop a low-complexity channel matrix completion approach tailored for ELAA communication systems.

B. RELATED WORKS

1) CODEBOOK-BASED CHANNEL ESTIMATION

Codebook-based channel estimations were widely adopted in commercial massive multi-input-multi-output (MIMO) communication systems, seeing widespread application in 5G new air (NR) and IEEE 802.11ad standards [2], [3], [4], [5]. With the advent of 5G NR release 17, the Type-I and Type-II codebooks were proposed for single-user and multi-user MIMO systems, respectively, with Type-I codebooks capturing single beam information and Type-II codebooks expanding this to include beam angles for enhanced multi-user supports [4]. Advanced evolution of the Type-II codebook leverages the channel sparsity for pilot-size reduction and incorporates non-line-of-sight (NLOS) path considerations for accuracy improvement [5], [15]. In addition, recent advances have pivoted towards deep learning to overcome the inherent limitations of the traditional codebook-based channel estimation, which treats the channel matrix as an image for more nuanced channel recovery [16]. The autoencoder (AE) model was proposed to condense the channel matrix into a streamlined and low-dimensional vector [16]. Based on the low-dimensional representation of the channel matrix, the vector-quantization variational autoencoder was proposed to learn the optimal codebook adaptively [17], [18]. To enhance AE efficiency and curb the growth of parameters, a self-information model has been synergized into the data-driven AE approach [19]. Moreover, an anti-noise AE has been devised to bolster channel estimation's resilience to noise, aiming to preserve signal integrity at the receiver [20]. Nonetheless, the traditional codebook-based channel estimation concentrates on beam-level characteristics and cannot reconstruct the complete channel matrix. The success of deep learning in channel estimation heavily depends on the data quality and the neural network model's generalization, which is challenging for achieving uniformly low normalized mean square error (NMSE) in the fluctuating environment of ELAA systems.

2) MATRIX COMPLETION FOR CHANNEL ESTIMATION

Matrix completion, a concept widely explored in fields such as signal recovery, machine learning, and recommendation systems, seeks to reconstruct a target matrix from its partially observed counterpart, leveraging the intrinsic structure of the target matrix [12], [13], [21], [22], [23], [24], [25], [26], [27], [28]. Specifically, the challenge of low-rank matrix completion was commonly addressed through nuclear norm minimization to achieve the balance between accuracy and computational feasibility [23], [24]. Further enhancements for low-rank matrix completion were achieved by solving the nuclear norm regularized linear least squares problem using the subgradient descent algorithm [25]. Additionally, the matrix's sparsity is commonly quantified by the L_1 norm [12], [26]. To harness both the low-rankness and sparsity of the targe matrix, a joint nuclear and L_1 norm optimization problem was investigated, and a standard ADMM algorithm was proposed to solve this problem [13]. Note that the joint nuclear and L_1 norm optimization problem is a convex problem, the ADMM algorithm could converge to a global optimum in a finite number of iterations. Underpinning the assumption that each matrix row is drawn from a consistent vector distribution, an insight that has paved the way for applying optimal transport theory to matrix imputation, which focuses on minimizing the Sinkhorn distance of the partially observed matrix [27], [28].

Based on the advanced matrix completion techniques, imputation-based channel estimation techniques have significantly attracted a lot of antentions in the literature [29], [30], [31], [32], [33]. The vector approximate message passing (VAMP) technique has advanced channel coefficient estimation through statistical inference, which models the channel coefficient as a key parameter within the posterior distribution [29]. The orthogonal matching pursuit (OMP) algorithm has been widely employed for channel estimation by effectively identifying sparsity-inducing weights, offering a targeted solution for communication environments with inherently sparse channel structures [30]. Meanwhile, the singular value thresholding (SVT) algorithm was proposed for recovering low-rank channel matrices by dismissing singular values below specified thresholds [31]. Addressing the distribution characteristics of channel matrix singular values, the introduction of a trimming operation was proposed for concentrating on significant singular values [32]. A two-stage algorithm was proposed by integrating the VAMP and SVT algorithms, which aims to optimize the interplay between the low-rankness and sparsity of the channel matrix [33]. Although these algorithms have shown promising results in terms of estimation accuracy compared to the codebook-based techniques, they are computationally expensive in the ELAA systems due to the SVT operation for the large-size channel matrices.

C. CONTRIBUTIONS

This paper introduces a low-complexity matrix completion approach for ELAA channel estimations. The salient contributions of our paper are delineated below.

1) Based on the low-rankness of ELAA channel matrix and the sparsity of its beam representation, the channel matrix estimation is first formulated as a joint nuclear and L_1 norm optimization problem. Considering L_1 norm is non-differentiable, the proximal gradient descent (PGD) method is adopted to reformulate the

optimization problem, where the subgradient of nuclear norm is computed by the SVT of channel matrix and the proximal operator of L_1 norm is derived by the soft-thresholding operator.

2) For an ELAA communication system with N receiving antennas and a channel matrix of rank r, we propose a subspace-approximated (SA)-SVT-PGD algorithm to efficiently solve the formulated optimization problem, achieving a computational complexity of $O(Nr^2)$ per PGD iteration. Based on the subspace similarity of the channel's Gram matrix across consecutive PGD iterations, the SA-SVT-PGD algorithm enables subspace orthogonalization within SVT operations to be performed concurrently with PGD iterations, thereby eliminating the nested loop dedicated to subspace orthogonalization. We formally prove that as the PGD iterations converge, the approximated subspace asymptotically converges to the true subspace, ensuring both the accuracy and computational efficiency of the proposed SA-SVT-PGD algorithm.

3) Simulation results demonstrate that the proposed SA-SVT-PGD algorithm reduces the convergence time by a maximum of 71.7% compared to the SVT-PGD algorithm. Additionally, under a given coherence time constraint, the proposed SA-SVT-PGD algorithm can improve the lower bound of spectral efficiency by a maximum of 112% compared to the SVT-PGD algorithm.

D. NOTATIONS AND OUTLINE

The notations and abbreviations are summarized in Table 1. The rest paper is organized as follows. The system model and problem formulation are introduced in Section II. The proposed algorithm and the convergence proof are given in Section III. The simulation results are shown in Section IV. Finally, the conclusion is given in Section V.

II. SYSTEM MODEL AND PROBLEM FORMULATION

A. ANTENNA-DOMAIN AND BEAM-DOMAIN REPRESENTATION OF ELAA CHANNEL

As shown in Figure 1, consider a uniform panel ELAA system with $L = L_h \times L_v$ transmitting antennas and $N = N_h \times N_v$ receiving antennas, where L_h and L_v are the number of transmitting antennas at the horizontal and vertical directions, N_h and N_v are the number of receiving antennas at the horizontal and vertical directions. Denote λ as the wavelength of the ELAA system, d as the antenna spacing, θ_l^t, ϕ_l^t as the elevation and azimuth angles of the transmit antennas relative to the l-th propagation path, respectively. Then the array response vector of transmit antennas for the l-th propagation path can be represented by

$$\mathbf{a}_t\left(\theta_l^t, \phi_l^t\right) \in \mathbb{C}^L = \mathbf{a}_{t,v}\left(\theta_l^t, \phi_l^t\right) \otimes \mathbf{a}_{t,h}\left(\theta_l^t, \phi_l^t\right), \quad (1)$$

where $\otimes$ denotes the Kronecker product,

$$\mathbf{a}_{t,v}\left(\theta_l^t, \phi_l^t\right) \in \mathbb{C}^{L_v} = \left[1, e^{j\frac{2\pi}{\lambda}d\sin(\theta_l^t)}, \ldots, e^{j\frac{2\pi}{\lambda}(L_v-1)d\sin(\theta_l^t)}\right]^T, \quad (2)$$

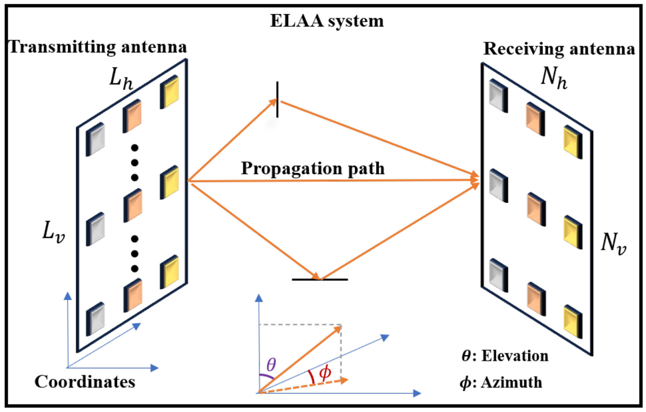


FIGURE 1. Diagram of the ELAA system.

and

$$\mathbf{a}_{t,h}\left(\theta_l^t, \phi_l^t\right) \in \mathbb{C}^{L_h} = \left[1, e^{j\frac{2\pi}{\lambda}d\sin(\phi_l^t)\cos(\theta_l^t)}, \ldots, e^{j\frac{2\pi}{\lambda}(L_h-1)d\sin(\phi_l^t)\cos(\theta_l^t)}\right]^T. \quad (3)$$

Similarly, the array response vector of receiving antennas for the l-th propagation path can be represented by

$$\mathbf{a}_r\left(\theta_l^r, \phi_l^r\right) \in \mathbb{C}^N = \mathbf{a}_{r,v}\left(\theta_l^r, \phi_l^r\right) \otimes \mathbf{a}_{r,h}\left(\theta_l^r, \phi_l^r\right), \quad (4)$$

where θ_l^r and ϕ_l^r are the elevation and azimuth angles of the receiving antennas relative to the l-th propagation path, respectively,

$$\mathbf{a}_{r,v}\left(\theta_l^r, \phi_l^r\right) \in \mathbb{C}^{N_v} = \left[1, e^{j\frac{2\pi}{\lambda}d\sin(\theta_l^r)}, \ldots, e^{j\frac{2\pi}{\lambda}(N_v-1)d\sin(\theta_l^r)}\right]^T, \quad (5)$$

and

$$\mathbf{a}_{r,h}\left(\theta_l^r, \phi_l^r\right) \in \mathbb{C}^{N_h} = \left[1, e^{j\frac{2\pi}{\lambda}d\sin(\phi_l^r)\cos(\theta_l^r)}, \ldots, e^{j\frac{2\pi}{\lambda}(N_h-1)\sin d(\phi_l^r)\cos(\theta_l^r)}\right]^T. \quad (6)$$

Based on the representation of the array response vectors, the geometric model of the ELAA channel can be represented by [34]

$$\mathbf{H} \in \mathbb{C}^{N\times L} = \sum_{l=1}^{\mathcal{L}} \alpha_l \mathbf{a}_r\left(\theta_l^r, \phi_l^r\right)\mathbf{a}_t^H\left(\theta_l^t, \phi_l^t\right), \quad (7)$$

where $\mathcal{L}$ is the total number of propagation paths, α_l is the complex fading coefficient of the l-th path. Since each element of $\mathbf{H}$ characterizes the channel fading coefficients at responding antennas, (7) is termed as the antenna-domain representation of the ELAA channel.

The antenna-domain representation of the ELAA channel can be further transformed into the beam-domain representation. Denote $[\theta_{\min}^t, \theta_{\max}^t]$ and $[\phi_{\min}^t, \phi_{\max}^t]$ as the elevation and azimuth angle range for transmissions, respectively. Denote the discrete transmission angle grids for the elevation and azimuth directions as

$$\theta^t[n_v] = \theta_{\min}^t + n_v\frac{\theta_{\max}^t - \theta_{\min}^t}{L_v - 1}, n_v \in [0, 1, \ldots, L_v - 1], \quad (8)$$

TABLE 1. Notations and abbreviations.

Notations	meaning	Notations	meaning
$(\cdot)^T$	Matrix transpose	$(\cdot)^H$	Matrix conjugate transpose
$(\cdot)^*$	Matrix conjugate	tr	Matrix trace
$(\cdot)^{-1}$	Matrix inverse	$\mathbf{I}_N$	$N \times N$ identity matrix
$\mathbb{O}_{N\times M}$	$N \times M$ zero matrix	**det**	Matrix determinant
vec	Vectorization operator	**unvec**	Un-vectorization operator
diag	Matrix diagonalization operator	∇	Gradient operator
$\Re$	Real part of a complex number	$\Im$	Imaginary part of a complex number
$sign$	Sign function	$\log_2$	Logarithm function with base 2
$\otimes$	Kronecker product	$\odot$	Hadamard product
$[x_i]_i \in \{0, 1, \ldots, r\}$	sequence of $\{x_0, \ldots, x_r\}$	$\|\|\cdot\|\|_*$	Nuclear norm
$\|\|\cdot\|\|_1$	Matrix L_1 norm	$\|\|\cdot\|\|_F$	Matrix Frobenius norm
Abbreviation	Full Name	Abbreviation	Full Name
ELAA	Extremely Large Antenna Array	SVT	Singular Value Thresholding
PGD	Proximal Gradient Descent	SA	Subspace-Approximated
VAMP	Vector Approximate Message Passing	AE	Auto-Encoder
OMP	Orthogonal Matching Pursuit	ADMM	Alternating Direction Method of Multipliers
NN	Neural Network	NMSE	Normalized Mean Square Error
ASE	Achieved Spectral Efficiency	DFT	Discrete Fourier Transform

and

$$\phi^t[n_h] = \phi^t_{\min} + n_h \frac{\phi^t_{\max} - \phi^t_{\min}}{L_h - 1}, n_h \in [0, 1, \ldots, L_h - 1]. \quad (9)$$

Then, the transforming matrix from antenna-domain to beam-domain for transmitting antennas is represented by

$$D_t \in \mathbb{C}^{L\times L} = \big[\mathbf{a}_t \ \ \big(\theta^t[0]\big), \phi^t[0]), \ldots, \mathbf{a}_t\big(\theta^t[L_v - 1]\big), \phi^t[L_h - 1])\big]. \quad (10)$$

Similarly, denote $[\theta^r_{\min}, \theta^r_{\max}]$ and $[\phi^r_{\min}, \phi^r_{\max}]$ as the elevation and azimuth angle range for receiving. Denote the discrete arriving angle grids for the elevation and azimuth directions as

$$\theta^r[m_v] = \theta^r_{\min} + m_v \frac{\theta^r_{\max} - \theta^r_{\min}}{N_v - 1}, m_v \in [0, 1, \ldots, N_v - 1], \quad (11)$$

and

$$\phi^r[m_h] = \phi^r_{\min} + m_h \frac{\phi^r_{\max} - \phi^r_{\min}}{N_h - 1}, m_h \in [0, 1, \ldots, N_h - 1] \quad (12)$$

Then, the transforming matrix from antenna-domain to beam-domain for receiving antennas is represented by

$$D_r \in \mathbb{C}^{N\times N} = \big[\mathbf{a}_r\big(\theta^r[0]\big), \phi^r[0]), \ldots, \mathbf{a}_r\big(\theta^r[N_v - 1]\big), \phi^r[N_h - 1])\big]. \quad (13)$$

Then the beam-domain representation of $\mathbf{H}$ is expressed by

$$Z = D_r^H \mathbf{H} D_t. \quad (14)$$

Note that each element of Z gives the virtual gain at the corresponding $\theta^r, \theta^t, \phi^r, \phi^t$.

B. LOW-RANKNESS AND SPARSITY OF ELAA CHANNEL

Based on the antenna-domain representation of ELAA channel, the channel matrix $\mathbf{H}$ could be represented by

$$\mathbf{H} = \mathbf{A_r} \mathbf{P} \mathbf{A}_t^H, \quad (15)$$

where $\mathbf{A}_r = [\mathbf{a}_r(\theta^r_1, \phi^r_1), \ldots, \mathbf{a}_r(\theta^r_{\mathcal{L}}, \phi^r_{\mathcal{L}})]$ and $\mathbf{A}_t^H = [\mathbf{a}_t^H(\theta^t_1, \phi^t_1), \ldots, \mathbf{a}_t^H(\theta^t_{\mathcal{L}}, \phi^t_{\mathcal{L}})]$, $\mathbf{P} = \mathbf{diag}(\alpha_1, \ldots, \alpha_L)$. Based on the rank property, i.e.,

$$rank(\mathbf{H}) \leq \min\{rank(\mathbf{A}_r) = rank(\mathbf{P}) = rank(\mathbf{A}_t) = \mathcal{L}\}, \quad (16)$$

then the rank of $\mathbf{H}$ is less than or equal to $\mathcal{L}$. In the ELAA system, the number of propagation paths $\mathcal{L}$ is significantly less than L and N. Therefore, the low-rankness is exhibited for $\mathbf{H}$.

Combining the equations (14) and (15), Z can be further represented by

$$Z = D_r^H \mathbf{H} D_t = D_r^H \mathbf{A_r} \mathbf{P} \mathbf{A}_t^H D_t. \quad (17)$$

Since only a limited number of propagation paths exist in the ELAA channel, $D_r\mathbf{A_r}$ and $D_t\mathbf{A_t}$ are the matrix where many elements are near-zero. Due to $\mathbf{P}$ is a diagonal matrix, the matrix Z also has a lot of near-zero elements.

C. INCOMPLETENESS OF CHANNEL MATRIX

Noth that the practical mobile communication system is a broadband system with a large number of subcarriers. To eliminate the interference among the subcarriers, the OFDM technique is adopted. Figure 2 illustrates a channel estimation diagram in an OFDM grid. Before transmitting the data symbols, the pilot symbols are first transmitted to estimate the channel matrix. A row of the channel matrix can be estimated per each pilot symbol. For a typical pilot configuration scheme in 5G 38.211 specifications [37], only

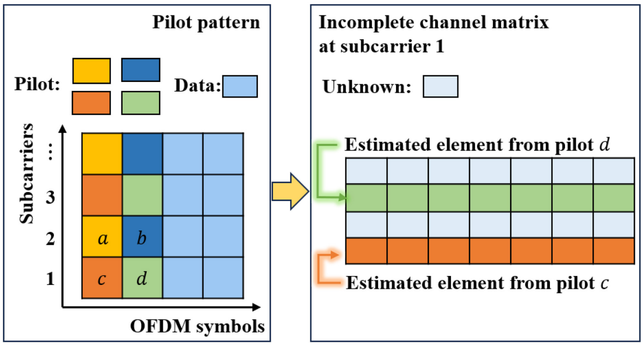


FIGURE 2. Diagram of the channel estimation and pilot allocation.

four pilot symbols are allocated per 14 OFDM symbols. Denote OF as the number of OFDM symbols in one coherence time, where the ELAA channel is assumed to be constant. Then $\frac{OF}{14} \times 4$ are the maximum permitted number of pilot symbols within one coherence time. To obtain the complete channel matrix at a subcarrier for the $N \times L$ ELAA channel matrix, N pilot symbols are required at a subcarrier. Hence, $N - \frac{OF}{14} \times 4$ is the number of rows where the element is unknown. Consider a classical ELAA configuration with $N = 100$ and $OF = 140$, only 40% row of the ELAA channel matrix can be estimated. In this paper, denote the missing rate as

$$\eta_m = 1 - \frac{OF}{14N} \times 4, \tag{18}$$

which characterizes the ratio of the number of missing rows to the total number of rows in the ELAA channel matrix.

D. PROBLEM FORMULATION

This paper aims to estimate the missing element of the ELAA channel matrix based on the low-rankness of $\mathbf{H}$ and the sparsity of Z. Denote the measured channel matrix as $\mathbf{H}^{\Omega} = P_{\Omega}(\mathbf{H})$, where Ω is the index set of the known channel elements and $P_{\Omega}(\cdot)$ is the projection operator that selectively zeroes out unknown elements, defined by the Hadamard product: $P_{\Omega}(\mathbf{H}) = \mathbf{H} \odot \mathbf{I}_{\Omega}$, with $\mathbf{I}_{\Omega}(i,j) = 1$ if $(i,j) \in \Omega$; otherwise, $\mathbf{I}_{\Omega}(i,j) = 0$. Addressing the challenges posed by the non-convex and non-smooth nature of the rank function, we adopt the nuclear norm as a surrogate to approximate the matrix rank. The nuclear norm, $\|\mathbf{H}\|_*$, is defined as the sum of the singular values of $\mathbf{H}$

$$\|\mathbf{H}\|_* = \sum_{i=1}^{\min(N,L)} \sigma_i(\mathbf{H}), \tag{19}$$

where $\sigma_i(\mathbf{H})$ represents the i-th singular value of $\mathbf{H}$. The sparsity of a matrix is quantified using the L_1 norm [35], which is expressed as

$$\|Z\|_1 = \max_j \sum_i |z_{i,j}|, \tag{20}$$

with $z_{i,j}$ being the (i, j)-th element of Z. Then, the ELAA channel completion is formulated as a constrained nuclear and L_1 norm minimization problem, which is represented by

Problem 1

$$\begin{aligned} \min_{\mathbf{H}} \;& \lambda_1 \|\mathbf{H}\|_* + \lambda_2 \|Z\|_1, \\ \text{s.t.} \;& P_{\Omega}(\mathbf{H}) = P_{\Omega}(\mathbf{M}), \end{aligned} \tag{21}$$

where λ_1 and λ_2 are regularization parameters that balance the minimization of the nuclear norm and the L_1 norm, respectively, $\mathbf{M}$ represents the true and complete ELAA channel matrix.

E. REVISITING THE ADMM ALGORITHM FOR PROBLEM 1

Since the nuclear norm and the L_1 norm are convex functions, Problem 1 constitutes a convex optimization problem with affine constraints, making it amenable to solution via the ADMM algorithm [13], [38]. To facilitate the application of the ADMM algorithm, Problem 1 is restructured as

Problem 2

$$\begin{aligned} \min_{\mathbf{H},Z,X,Y} \;& \lambda_1 \|\mathbf{H}\|_* + \lambda_2 \|Z\|_1 + \frac{1}{2}\|X\|_F^2 \\ & + \frac{1}{2}\|P_{\Omega}(Y) - P_{\Omega}(\mathbf{M})\|_F^2, \\ \text{s.t.} \;& \mathbf{H} = Y, \quad X = Y - D_r Z D_t^H, \end{aligned} \tag{22}$$

where Y and $X = Y - D_r Z D_t^H$ are introduced as auxiliary matrix variables to decompose the original problem. Problem 2 is transformed into an unconstrained problem by the augmented Lagrangian multiplier method, which is expressed by

$$\begin{aligned} \min_{\{\mathbf{H},Z,X,Y,Z_1,Z_2\}} \;& \lambda_1 \|\mathbf{H}\|_* + \lambda_2 \|Z\|_1 + \frac{1}{2}\|X\|_F^2 \\ & + \frac{1}{2}\|P_{\Omega}(Y) - P_{\Omega}(\mathbf{M})\|_F^2 \\ & + \mathrm{tr}\Big(Z_1^H(\mathbf{H} - Y)\Big) + \frac{\rho}{2}\|\mathbf{H} - Y\|_F^2 \\ & + \mathrm{tr}\Big(Z_2^H\Big(Y - D_r Z D_t^H - X\Big)\Big) \\ & + \frac{\rho}{2}\|Y - D_r Z D_t^H - X\|_F^2, \end{aligned} \tag{23}$$

where ρ is the stepsize parameter of the ADMM algorithm, and Z_1 and Z_2 are the Lagrange multipliers. The ADMM algorithm to solve (23) is summarized as Algorithm 1.

Remark 1: The key computational steps within each iteration of Algorithm 1 are the SVD of $Y^{(k)} - \frac{1}{\rho}Z_1^{(k)}$ and the computation of $Y^{(k+1)}$, i.e., equation (24) and (26). The computation complextiy of the SVD of $Y^{(k)} - \frac{1}{\rho}Z_1^{(k)}$ is $O(N^2L)$ and the computation complexity of $Y^{(k+1)}$ is $O(N^2L^2)$. When the number of transmitting antennas exceeds one thousand in ELAA systems, this level of complexity renders the direct application of Algorithm 1 impractical within the constraints of real-time processing. In addition, as shown in equation (26), the matrix $A^HA + 2\rho I$ is required to be stored in the memory and its inverse should be

Algorithm 1 ADMM Algorithm for Problem 2

Require:

Initial guesses: $\mathbf{H}^{(0)}$, $Z^{(0)}$, $X^{(0)}$, $Y^{(0)}$, $Z_1^{(0)}$, $Z_2^{(0)}$;

Parameters: ρ, λ_1, λ_2;

Known matrix: Ω, $P_\Omega(\mathbf{M})$, D_r, D_t, $A \triangleq \sum_{i=1}^{N} \text{diag}(\Omega_{i:})^T \otimes \mathbf{I}_N$, $B = D_t^* \otimes D_r$.

1: **repeat**

2: Update $\mathbf{H}^{(k+1)}$ by

$$U\text{diag}([\sigma_i]_i)V^H = SVD\left(Y^{(k)} - \frac{1}{\rho}Z_1^{(k)}\right), \tag{24}$$

where σ_i, U, and V denote the i-th singular value, and the left and right singular vectors of the matrix $Y^{(k)} - \frac{1}{\rho}Z_1^{(k)}$, respectively.

$$\mathbf{H}^{(k+1)} = U\text{diag}\big(\{\max(\sigma_i - \lambda_1/\rho, 0)\}_{1\le i\le r}\big)V^H, \tag{25}$$

where r is the rank of matrix $Y^{(k)} - \frac{1}{\rho}Z_1^{(k)}$.

3: Update $Y^{(k+1)}$ by

$$Y^{(k+1)} = \text{unvec}\Big(\left(A^H A + 2\rho\mathbf{I}\right)^{-1}\left(\mathbf{z}_1^{(k)} + \rho\mathbf{h}^{(k+1)} + A^H\mathbf{h}_\Omega + \mathbf{z}_2^{(k)} + \rho\mathbf{x}^{(k)} + \rho B\mathbf{z}^{(k)}\right)\Big), \tag{26}$$

where $\mathbf{z}_1^{(k)} = \text{vec}(Z_1^{(k)})$, $\mathbf{z}_2^{(k)} = \text{vec}(Z_2^{(k)})$, $\mathbf{x}^{(k)} = \text{vec}(X^{(k)})$, $\mathbf{z}^{(k)} = \text{vec}(Z^{(k)})$, $\mathbf{h}^{(k)} = \text{vec}(\mathbf{H}^{(k)})$, $\mathbf{h}_\Omega = \text{vec}(P_\Omega(\mathbf{M}))$.

4: Update $Z^{(k+1)}$ by

$$\begin{aligned}\hat{\mathbf{z}}^{(k+1)} &\triangleq \text{unvec}\left(D_r^H\left(\frac{1}{\rho}Z_2^{(k)} - X^{(k)} + Y^{(k+1)}\right)D_t\right),\\ Z^{(k+1)} &= \text{unvec}\left(\max\left(\left|\Re\left(\hat{\mathbf{z}}^{(k+1)}\right)\right| - \frac{\lambda_2}{\rho}, 0\right) + j\max\left(\left|\Im\left(\hat{\mathbf{z}}^{(k+1)}\right)\right| - \frac{\lambda_2}{\rho}, 0\right)\right).\end{aligned} \tag{27}$$

5: Update $X^{(k+1)}$ by

$$X^{(k+1)} = \frac{\rho}{\rho+1}\left(Y^{(k+1)} - D_r Z^{(k+1)} D_t^H + \frac{1}{\rho}Z_2^{(k)}\right). \tag{28}$$

6: Update $Z_1^{(k+1)}$ by

$$Z_1^{(k+1)} = Z_1^{(k)} + \rho\left(\mathbf{H}^{(k+1)} - Y^{(k+1)}\right). \tag{29}$$

7: Update $Z_2^{(k+1)}$ by

$$Z_2^{(k+1)} = Z_2^{(k)} + \rho\left(Y^{(k+1)} - D_r Z^{(k+1)} D_t^H - X^{(k+1)}\right). \tag{30}$$

8: **until** Convergence is achieved.

performed in Algorithm 1. For an ELAA communication system with $L = 1024$ and $N = 256$, the required memory would reach 512 Giga-Bytes, which significantly exceeds the practical memory in the communication system. Although Algorithm 1 exhibits high computational and memory overhead, it inspires this paper to optimize the nuclear norm using the SVD operator as (24) and optimize the L_1 norm as (27).

III. PROPOSED SUBSPACE-APPROXIMATED SVT-PGD ALGORITHM

In this section, the PGD framework is first adopted to solve the nuclear and L_1 norm optimization problem. Based on the PGD framework, a subspace-approximated SVT-PGD algorithm is proposed to reduce the computational complexity of SVT operation.

A. PRELIMINARIES OF THE PROXIMAL GRADIENT DESCENT ALGORITHM

The PGD algorithm stands apart from the ADMM algorithm by avoiding the use of auxiliary variables. It excels in optimizing composite objective functions without necessitating the differentiability of the objective function. Given a composite function $f(x) = g(x) + h(x)$, where $g(x)$ is a differentiable convex function and $h(x)$ represents a non-differentiable convex function, and x is within a bounded closed set, the PGD algorithm seeks to minimize $f(x)$ through iterative updates. The updated formula for PGD is

$$x^{(k+1)} := \mathbf{prox}_{t,h}\Big(x^{(k)} - t\nabla g\Big(x^{(k)}\Big)\Big), \tag{31}$$

where is equivalent to

$$x^{(k+1)} = \underset{z}{\operatorname{argmin}}\ \frac{1}{2t}\|z - \Big(x^{(k)} - t\nabla g(x^{(k)})\Big)\|_2^2 + h(z), \tag{32}$$

where t represents the stepsize, k is the iteration count, and **prox** symbolizes the proximal mapping operator. This operator is designed to balance the trade-off between adhering closely to the gradient step $x - t\nabla g(x)$ and minimizing the function $h(z)$. When the optimization error ϵ is given, the PGD algorithm is guaranteed to converge to the optimal solution of the composite function $f(\cdot)$, boasting a convergence rate of $O(1/\sqrt{\epsilon})$ [39].

B. THE PROPOSED SVT-PGD ALGORITHM FOR ELAA CHANNEL MATRIX COMPLETION

Transforming Problem 1 into an unconstrained optimization by penalty function method, we get

Problem 3

$$f(\mathbf{H}) := \lambda_1\|\mathbf{H}\|_* + \frac{1}{2t}\|P_\Omega(\mathbf{H}) - P_\Omega(\mathbf{M})\|_F^2 + \lambda_2\|D_r^H\mathbf{H}D_t\|_1. \tag{33}$$

Denotes $g(\mathbf{H})$ as

$$g(\mathbf{H}) := \frac{1}{2t}\|P_\Omega(\mathbf{H}) - P_\Omega(\mathbf{M})\|_F^2 + \lambda_1\|\mathbf{H}\|_*, \tag{34}$$

and $h(\mathbf{H})$ as

$$h(\mathbf{H}) = \lambda_2\|D_r^H\mathbf{H}D_t\|_1. \tag{35}$$

The proximal mapping for $f(\mathbf{H})$ then becomes

$$\mathbf{H}^{(k+1)} := \underset{J}{\operatorname{argmin}}\ \frac{1}{2t}\|J - \Big(\mathbf{H}^{(k)} - t\nabla g(\mathbf{H}^{(k)})\Big)\|_F^2 + \lambda_2\|D_r^HJD_t\|_1. \tag{36}$$

Although $\|\mathbf{H}\|_*$ is non-differentiable as $\mathbf{H}$ has repetitive singular values, but $\mathbf{H}^{(k)} - t\nabla g(\mathbf{H}^{(k)})$ could be approximated by its sub-differential, i.e.,

$$\mathbf{H}^{(k)} - t\nabla g\Big(\mathbf{H}^{(k)}\Big) \approx U^k \operatorname{diag}\Big(\{\max\Big(\sigma_i^k - \frac{\lambda_1}{t}, 0\Big)\}_{1\le i\le r}\Big)V^{kH}. \tag{37}$$

where U^k and V^k are the left and right singular matrix of $\mathbf{H}^{(k)}$, respectively, and σ_i^k denotes the i-th singular value of $\mathbf{H}^{(k)}$. Based on (37), (36) can be simplified

$$\mathbf{H}^{(k+1)} = \underset{J}{\operatorname{argmin}}\ \frac{1}{2t}\|J - U^kD^kV^{kH}\|_F^2 + \lambda_2\|D_r^HJD_t\|_1, \tag{38}$$

where $D^k = \operatorname{diag}(\{\max(\sigma_i^k - \frac{\lambda_1}{t}, 0)\}_{1\le i\le r})$. When D^k is given, (38) is reduced to solving for $Z^{(k+1)}$ in the beam domain:

$$Z^{(k+1)} := \underset{Z}{\operatorname{argmin}}\ \frac{1}{2t}\|Z - D_r^HU^kD^kV^{kH}D_t\|_F^2 + \lambda_2\|Z\|_1,$$
$$\mathbf{H}^{(k+1)} = D_rZ^{(k+1)}D_t^H. \tag{39}$$

Algorithm 2 SVT-PGD Algorithm for ELAA Channel Matrix Completion

Require: Initial values $\mathbf{H}^{(0)}$, $Z^{(0)}$, stepsize t, regularization parameters λ_1, λ_2, observation set Ω, observed matrix $\mathbf{M}$, DFT matrices D_r, D_t.
1: **repeat**
2: Apply SVT to update $\mathbf{H}^{(k)} - t\nabla g(\mathbf{H}^{(k)})$ as (37);
3: Solve for $Z^{(k+1)}$ by (40);
4: Update $\mathbf{H}^{(k+1)}$ by computing $D_rZ^{(k+1)}D_t^H$.
5: Update $\mathbf{H}^{(k+1)}$ by computing $P_\Omega^\perp(\mathbf{H}^{(k+1)}) + P_\Omega(\mathbf{M})$.
6: **until** Convergence is reached.

Similar to (27) of ADMM, the closed-form solution for each element of $Z^{(k+1)}$ is

$$\Big[Z^{(k+1)}\Big]_{ij} = \operatorname{sign}\big(\Re(\beta_{ij})\big)\max\Big(|\Re(\beta_{ij})| - \frac{\lambda_2}{t}, 0\Big) + j\operatorname{sign}\big(\Im(\beta_{ij})\big)\max\Big(|\Im(\beta_{ij})| - \frac{\lambda_2}{t}, 0\Big), \tag{40}$$

where $\beta_{ij} = [D_r^HU^kD^kV^{kH}D_t]_{ij}$.

Remark 2: The computation of (37)-(40) is termed as SVT-PGD algorithm and summarized in Algorithm 2. Similar to the ADMM algorithm, each iteration of SVT-PGD algorithm has a closed-form expression. However, the SVT-PGD algorithm has saved the computational overhead for computing the auxiliary variable $Y^{(k+1)}$ in the ADMM algorithm. The key step in the SVT-PGD algorithm is the SVT operation for updating $\mathbf{H}^{(k)} - t\nabla g(\mathbf{H}^{(k)})$.

C. COMPUTATIONAL COMPLEXITY ANALYSIS OF SVT-PGD ALGORITHM

Note that the computation device in modern communication system has been empowered by multi-core general-purpose graphic processing units (GPGPU), which can efficiently perform matrix-vector multiplications and matrix products using a large number of computing cores. However, the current GPGPU architecture is still limited in computing the operation that involves large number of iteration, such as the eigenvalue decomposition (EVD) and QR decomposition. In fact, the EVD would occupy about 90% of the total computation time of the SVT-PGD algorithm in the ELAA communication system. Therefore, this subsection will focus on analyzing the computational complexity of SVT operation that is implemented using EVD.

Denote $Q_k = P_\Omega^\perp(\mathbf{H}^{(k)}) + P_\Omega(\mathbf{M}) \in \mathbb{C}^{N\times L}$, the traditional SVT computation with complexity is presented as follows.

1) Matrix product: Forming matrix $Q_kQ_k^H \in \mathbb{C}^{N\times N}$ with complexity of $O(N^2L/G)$, G is the number of parallel computing cores;
2) Computing left singular matrix $U = [u_1, u_2, \ldots, u_r] \in \mathbb{C}^{N\times r}$: Performing eigen-decomposition (EVD) on $Q_kQ_k^H$ with complexity of $O(N^3)$;
3) Computing right singular matrix $V = [v_1, v_2, \ldots, v_r] \in \mathbb{C}^{L\times r}$: Taking the square root of r eigenvalues of $Q_kQ_k^H$

Algorithm 3 Subspace Orthogonal Iteration (SOI) Algorithm

Initialize a rank r matrix W for EVD. Initialize an random $N \times r$ column orthogonal matrix $O^{(0)}$.
Set $k = 0$.
while not converged **do**
  Compute $Y_{k+1} = WO^{(k)}$.
  Perform QR decomposition: $Y_{k+1} = O^{(k+1)}R^{(k+1)}$.
  Set $k = k + 1$.
end while
return $O^{(k)}$ as the output.

with complexity of $O(r/G)$. Forming matrix U_k by stacking $v_i = \frac{Q_k^H u_i}{\sigma_i}, i = 1, 2, \ldots, r$ with complexity of $O(\frac{r}{G}(NL + r))$;

The total complexity of traditional SVT computation is $O(N^3 + \frac{(N^2L+rNL+r^2+r)}{G})$. Obiously, the EVD computation dominates the total computation complexity of traditional SVT.

Since $Rank(Q_kQ_k^H) = Rank(Q_k)$, $Q_kQ_k^H$ is still a low-rank matrix whose EVD can be computed efficiently by subspace orthogonal iteration algorithm [35]. Denote $W = Q_kQ_k^H \in \mathbb{C}^{N\times N}$, then its EVD could be represented as

$$W = \begin{bmatrix} \tilde{V} & V_\perp \end{bmatrix} \begin{bmatrix} \Lambda_r & \mathbb{O} \\ \mathbb{O} & \mathbb{O} \end{bmatrix} \begin{bmatrix} \tilde{V}^H \\ V_\perp^H \end{bmatrix}, \quad (41)$$

with $\tilde{V} \in \mathbb{C}^{N\times r}$ encompassing the significant eigenvectors of W, $V_\perp \in \mathbb{C}^{N\times(N-r)}$ covering the remaining eigenvectors, and $\Lambda_r \in \mathbb{R}^{r\times r}$ representing the principal eigenvalues. The following lemma presents the relationship between $\tilde{V}$ and the column space of W.

Lemma 1 [35]: Consider a rank-r matrix $W \in \mathbb{C}^{N\times N}$ with eigenmatrix $\tilde{V}$, where $r \ll N$. Let $O \in \mathbb{C}^{N\times r}$ represent a column orthogonal matrix spanning the column space of W. It follows from the relationship between the column space and the eigenspace that $\tilde{V} = OY$, where Y is the eigenmatrix of O^HWO.

Lemma 1 reveals a crucial insight: if a column orthogonal matrix O the low-rank matrix W is predetermined, we can efficiently obtain the eigenmatrix of W by OY, where Y is the eigenmatrix of O^HWO. Note that O^HWO is a $r \times r$ matrix, whose computational complexity is reduced to $O(r^3)$. Nonetheless, the challenge in practical applications stems from the fact that O is not readily known a priori. To address this, the subspace orthogonal iteration (SOI) techniques are employed to approximate O. The computation process of SOI is summarized in Algorithm 3. The SOI-based SVT-PGD algorithm is presented in Algorithm 4.

The complexity of SOI-based EVD computation is presented as follows.

1) SOI for W: Performing Algorithm 3 to output a approximate of O, denoted as $\bar{O}$. In Algorithm 3, the computational complexity of QR decomposition is $O(r^2N)$, the computational complexity of the matrix product is $O(N^2r/G)$. Then the total complexity of Algorithm 3 is $O(M(r^2N + N^2r/G))$, where M is the iteration count of subspace orthogonal iterations.
2) EVD for $\bar{O}^HW\bar{O}$: Performing the EVD of $\bar{O}^HW\bar{O}$ to obtain its eigenmatrix Y. The matrix product of $\bar{O}^HW\bar{O}$ has a computational complexity of $O((N^2r+r^2N)/G)$. Since O^HWO is a $r \times r$ matrix, the computational complexity of its EVD is $O(r^3)$.
3) Matrix product for $\tilde{V}$: Compute $\tilde{V} = \bar{O}Y$. The matrix product has a computational complexity of $O((r^2N)/G)$.

Algorithm 4 SOI-SVT-PGD Algorithm

Require: Initial guesses for $\mathbf{H}^{(0)}$ and $Z^{(0)}$, step size t, regularization parameters λ_1 and λ_2, observation set Ω, observed matrix $\mathbf{M}$, DFT matrices D_r and D_t.
repeat
  Compute $Q_k = P_\Omega^\perp(\mathbf{H}^{(k)}) + P_\Omega(\mathbf{M})$, and $W_k = Q_kQ_k^H$.
  Use Algorithm 3 to compute the approximated column orthogonal matrix $\bar{O}_k$ for W_k;
  $[Y_k, \Lambda_k] = EVD(\bar{O}_k^HW_k\bar{O}_k)$ and $U_k = \bar{O}_kY_k$;
  $\Lambda_k^{1/2} = \Lambda_k^{1/2} - \frac{\lambda_1}{t}\mathbf{I}$;
  $V_k = \frac{Q_k^HU_k}{\Lambda_k^{1/2}}$;
  Update $Z^{(k+1)}$ by (40);
  Update $\mathbf{H}^{(k+1)}$ as $D_rZ^{(k+1)}D_t^H$;
  Update $\mathbf{H}^{(k+1)}$ by computing $P_\Omega^\perp(\mathbf{H}^{(k+1)}) + P_\Omega(\mathbf{M})$;
until convergence is achieved;

In total, the computation complexity of SOI-based EVD computation is $O(Mr^2N + \frac{(M+1)N^2r+2r^2N}{G} + r^3)$. Note that the number of computation core G would exceed one thousand in modern GPGPU systems. Therefore, multiple QR decompositions in SOI would dominates the total computation complexity. The complexity of $O(Mr^2N)$ is introduced due to the nested loop in the SVT operation.

D. SUBSPACE-APPROXIMATED SVT-PGD ALGORITHM

As shown in Algorithm 4, the computational complexity of the SOI-SVT-PGD algorithm is from iteratively approximating the column subspaces for each PGD iteration, which induces a nested loop structure. To cancel the nested loop structure, we propose a novel subspace-approximated SVT-PGD algorithm based on the similarity of the column subspaces across successive iterations in the SVT-PGD algorithm, which is summarized in Algorithm 5.

The distinctions between Algorithm 4 and Algorithm 5 are illustrated in Figure 3. As shown on line 7 of Algorithm 5, only once SOI iteration is performed to update the column subspaces of W_k in each PGD iterations. Besides matrix products, the computational complexity of Algorithm 5 is $O(r^3 + Nr^2)$ per iteration, which is much lower than the $O(r^3 + Mr^2N)$ complexity of Algorithm 4.

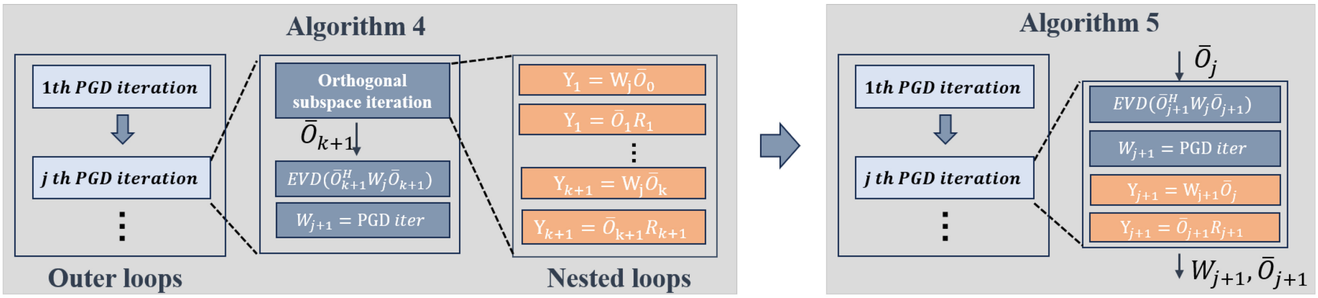


FIGURE 3. The difference between Algorithm 4 and Algorithm 5.

IV. CONVERGENCE ANALYSIS OF THE SUBSPACE-APPROXIMATED SVT-PGD ALGORITHM

A. FIX POINT ITERATION REPRESENTATION OF THE SUBSPACE-APPROXIMATED SVT-PGD ALGORITHM

The matrix W_k in Algorithm 5 is represented as a weighted sum of symmetric matrices, $W_k = \sum_{ij} B_{ij} W_{ij,k}$, where each B_{ij} is a matrix with zeroes everywhere except for ones at the (i, j)-th and (j, i)-th entries, and $W_{ij,k}$ represents the (i, j)-th entry of W_k. Given that B_{ij} matrices are predefined, we can express W_k as a function of its vectorized form x_k, i.e., $W_k = \mathcal{B}(x_k)$.

Let's denote the rank of W_k as r, and the column subspace of W_k as O_k. The EVD of W_k can be effectively achieved through the EVD of its orthogonal projection onto O_k, denoted by:

$$P_{O_k}(W_k) = \underset{\text{Range}(Y)=\text{Range}(O_k)}{\text{argmin}} \|Y - W_k\|_F^2 = O_k O_k^H W_k O_k O_k^H. \quad (42)$$

Following Lemma 1, we attain the EVD of $P_{O_k}(W_k)$ using $O_k Y_k$, where Y_k is the eigenmatrix of $O_k^H W_k O_k$. We define the function $\phi(\cdot)$ as:

$$\phi\left(P_{O_k}(W_k), P_{\hat{O}_k}(W_k^H)\right) = U^k \max\left(\sigma_i^k - \frac{\lambda_1}{t}, 0\right) V^{kH}, \quad (43)$$

where $V^k = \hat{O}_k \hat{Y}_k$, $U^k = O_k Y_k$, and σ_i^k is the i-th singular value of W_k, $\hat{Y}_k$ and $\hat{O}_k$ is the eigenmatrix and column orthogonal matrix of W_k^H, respectively. Thus, lines 2-3 of Algorithm 5 can be encapsulated by $\phi(P_{O_k}(W_k), P_{\hat{O}_k}(W_k^H))$. The function $\phi(\cdot)$ is Lipschitz continuous.

$$\|\phi(A, C) - \phi(B, C)\| \leq L_\phi \|A - B\|_F, \quad \forall A, B, C \in \mathbb{S}^n. \quad (44)$$

The subsequent iteration steps in Algorithm 5 can be represented as:

$$W_{k+1} = T\Big(\phi(P_{O_k}(W_k), P_{\hat{O}_k}(W_k^H))\Big), \quad (45)$$

where $T(\cdot)$ is also Lipschitz continuous with Lipschitz constant L_T. This leads to a reformulation of Algorithm 5's iteration process:

$$x_{k+1} = T\Big(\phi(P_{O_k}(\mathcal{B}(x_k)), P_{\hat{O}_k}(\mathcal{B}(x_k)^H))\Big), k = 0, 1, 2, \ldots \quad (46)$$

with the orthogonal updates:

$$O_{k+1} = \text{orth}(\mathcal{B}(x_{k+1})O_k), \quad k = 0, 1, 2, \ldots, \quad (47)$$

Algorithm 5 Subspace-Approximated SVT-PGD Algorithm

Require: Initial guesses: $\mathbf{H}^{(0)}$, $Z^{(0)}$; Step size: t; Regularization parameters: λ_1, λ_2; Observation set: Ω; Observed matrix: $\mathbf{M}$; DFT matrices: D_r, D_t; Initialize $Q_0 = P_\Omega^\perp(\mathbf{H}^{(0)}) + P_\Omega(\mathbf{M})$, $W_0 = Q_0^H Q_0$; Random initialization of $\bar{O}_0$;

1: **repeat**
2: $[Y_k, \Lambda_k] = EVD(\bar{O}_k^H W_k \bar{O}_k)$ and $U_k = \bar{O}_k Y_k$;
3: $\Lambda_k^{1/2} = \Lambda_k^{1/2} - \frac{\lambda_1}{t}\mathbf{I}$;
4: $V_k = \frac{Q_k^H U_k}{\Lambda_k^{1/2}}$;
5: Update $Z^{(k+1)}$ by (40);
6: Update $\mathbf{H}^{(k+1)}$ as $D_r Z^{(k+1)} D_t^H$.
7: Update $Q_{k+1} = P_\Omega^\perp(\mathbf{H}^{(k+1)}) + P_\Omega(\mathbf{M})$, $W_{k+1} = Q_{k+1}^H Q_{k+1}$.
8: Update $\bar{O}_{k+1}$ via QR decomposition of $W_{k+1}\bar{O}_k$.
9: Increment iteration count: $k = k + 1$.
10: **until** convergence is reached.

where "orth" signifies the operation of obtaining an orthogonal matrix, typically through QR decomposition.

B. CONVERGENCE PROOF OF THE SUBSPACE-APPROXIMATED SVT-PGD ALGORITHM

The cornerstone of our convergence analysis lies in demonstrating that the iterative process defined by (46)-(47) stabilizes to a fixed point. To this end, we first prove that the sequence generated by (46)-(47) converges to the sequence generated by $x_{k+1} = T(\phi(P_{X_k}(\mathcal{B}(x_k)), P_{\hat{X}_k}(\mathcal{B}(x_k)^H)))$, where X_k and $\hat{X}_k$ denote the true column subspaces of W_k and W_k^H, respectively. Then, we prove that the sequence generated by $x_{k+1} = T(\phi(P_{X_k}(\mathcal{B}(x_k)), P_{\hat{X}_k}(\mathcal{B}(x_k)^H)))$ converges to a fixed point.

Lemma 2: Let X_k and $\hat{X}_k$ represent the actual column subspaces of W_k and W_k^H, respectively. Similarly, let O_k and $\hat{O}_k$ be the approximated column subspaces of W_k and W_k^H, obtained through the iteration process outlined in (29)-(30). We assert that the sequence generated by $T(\phi(P_{O_k}(\mathcal{B}(x_k)), P_{\hat{O}_k}(\mathcal{B}(x_k)^H)))$ converges to the sequence produced by $T(\phi(P_{X_k}(\mathcal{B}(x_k)), P_{\hat{X}_k}(\mathcal{B}(x_k)^H)))$.

Proof: The error term between $\phi(P_{O^k}(\mathcal{B}(x_k)), P_{\hat{O}^k}(\mathcal{B}(x_k)))$ and $\phi(P_{X^k}(\mathcal{B}(x_k)), P_{\hat{X}^k}(\mathcal{B}(x_k)))$ can be calculated by

$$\begin{aligned}
&||\phi\Big(P_{O^k}(\mathcal{B}(x_k)), P_{\hat{O}^k}(\mathcal{B}(x_k))\Big) - \\
&\qquad \phi\Big(P_{X^k}(\mathcal{B}(x_k)), P_{\hat{X}^k}(\mathcal{B}(x_k))\Big)||_F \\
&= ||\phi\Big(P_{O^k}(\mathcal{B}(x_k)), P_{\hat{O}^k}(\mathcal{B}(x_k))\Big) \\
&\qquad -\phi\Big(P_{X^k}(\mathcal{B}(x_k)), P_{\hat{O}^k}(\mathcal{B}(x_k))\Big) \\
&\qquad +\phi\Big(P_{X^k}(\mathcal{B}(x_k)), P_{\hat{O}^k}(\mathcal{B}(x_k))\Big) \\
&\qquad -\phi\Big(P_{X^k}(\mathcal{B}(x_k)), P_{\hat{X}^k}(\mathcal{B}(x_k))\Big)||_F \\
&\leq ||\phi\Big(P_{O^k}(\mathcal{B}(x_k)), P_{\hat{O}^k}(\mathcal{B}(x_k))\Big) \\
&\qquad -\phi\Big(P_{X^k}(\mathcal{B}(x_k)), P_{\hat{O}^k}(\mathcal{B}(x_k))\Big)||_F \\
&\qquad +||\phi\Big(P_{X^k}(\mathcal{B}(x_k)), P_{\hat{O}^k}(\mathcal{B}(x_k))\Big) \\
&\qquad -\phi\Big(P_{X^k}(\mathcal{B}(x_k)), P_{\hat{X}^k}(\mathcal{B}(x_k))\Big)||_F \\
&\leq L_\phi ||P_{O^k}(\mathcal{B}(x_k)) - P_{X^k}(\mathcal{B}(x_k))||_F \\
&\qquad +L_\phi ||P_{\hat{O}^k}(\mathcal{B}(x_k)) - P_{\hat{X}^k}(\mathcal{B}(x_k))||_F, \qquad (48)
\end{aligned}$$

where the first inequality is due to the triangle inequality, the second inequality is due to the Lipschitz continuity of $\phi(\cdot)$. Furthermore, we have

$$\begin{aligned}
&||P_{O^k}(\mathcal{B}(x_k)) - P_{X^k}(\mathcal{B}(x_k))||_F \\
&= ||P_{O^k}(\mathcal{B}(x_k)) - O^k\Big(O^k\Big)^H \mathcal{B}(x_k) X^k \Big(X^k\Big)^H \\
&\quad + O^k\Big(O^k\Big)^H \mathcal{B}(x_k) X^k \Big(X^k\Big)^H - P_{X^k}(\mathcal{B}(x_k))||_F \\
&\leq 2||O^k\Big(O^k\Big)^H \mathcal{B}(x_k)\Big(O^k(O^k)^H - X^k(X^k)^H\Big)||_F \\
&\leq 2||O^k\Big(O^k\Big)^H||_F ||\mathcal{B}(x_k)||_F ||O^k\Big(O^k\Big)^H - X^k\Big(X^k\Big)^H||_F \\
&\leq 2BC||O^k\Big(O^k\Big)^H - X^k\Big(X^k\Big)^H||_F \qquad (49)
\end{aligned}$$

where B is the bound of the Frobenius norm of $\mathcal{B}(x_k)$, and C is the bound of the Frobenius norm of $||O^k(O^k)^H||_F$, the first inequality is due to the triangle inequality, the second inequality is due to $||AB|| \leq ||A||||B||$. As same as (49), we have

$$\begin{aligned}
&||P_{\hat{O}^k}(\mathcal{B}(x_k)) - P_{\hat{X}^k}(\mathcal{B}(x_k))||_F \\
&\leq 2||\hat{O}^k\Big(\hat{O}^k\Big)^H - \hat{X}^k\Big(\hat{X}^k\Big)^H||_F \\
&\leq 2BC||\hat{O}^k\Big(\hat{O}^k\Big)^H - \hat{X}^k\Big(\hat{X}^k\Big)^H||_F. \qquad (50)
\end{aligned}$$

Denote $X \in \mathbb{C}^{M\times N}$ and $Y \in \mathbb{C}^{M\times N}$ as the column orthogonal matrix. The singular value of $X^H Y$ is $\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_N \geq 0$. The principal angle between Range(X) and Range(Y) is defined as

$$\theta(X, Y) := \mathrm{diag}(\arccos\sigma_1, \ldots, \arccos\sigma_N). \qquad (51)$$

then we have

$$||\sin\theta(X, Y)||_F = ||XX^H - YY^H||_F. \qquad (52)$$

Based on the (50)-(52), we have

$$||P_{O^k}(\mathcal{B}(x_k)) - P_{X^k}(\mathcal{B}(x_k))||_F \leq 2BC||\sin\theta\Big(O^k, X^k\Big)||_F, \qquad (53)$$

and

$$||P_{\hat{O}^k}(\mathcal{B}(x_k)) - P_{\hat{X}^k}(\mathcal{B}(x_k))||_F \leq 2BC||\sin\theta\Big(\hat{O}^k, \hat{X}^k\Big)||_F. \qquad (54)$$

Due to the fact that O_k and $\hat{O}_k$ are obtained by orthogonal iterations, we have [35]

$$\begin{aligned}
&(a) \quad ||\sin\theta\Big(O^k, X^k\Big)||_F \leq \left(\frac{\gamma_{r+1}}{\gamma_r}\right)^k, \\
&(b) \quad ||\sin\theta\Big(\hat{O}^k, \hat{X}^k\Big)||_F \leq \left(\frac{\gamma_{r+1}}{\gamma_r}\right)^k, \qquad (55)
\end{aligned}$$

where γ_{r+1} is the $(r+1)$-th eigenvalue $\mathcal{B}(x_k)$. Due to the fact that $\mathcal{B}(x_k)$ is a low-rank matrix, we have $(\frac{\gamma_{r+1}}{\gamma_r})^k << 1$. With the increase of iteration steps, $(\frac{\gamma_{r+1}}{\gamma_r})^k \to 0$, thus

$$\begin{aligned}
&||\phi\Big(P_{O^k}(\mathcal{B}(x_k)), P_{\hat{O}^k}(\mathcal{B}(x_k))\Big) \\
&\quad -\phi\Big(P_{X^k}(\mathcal{B}(x_k)), P_{\hat{X}^k}(\mathcal{B}(x_k))\Big)||_F \qquad (56)
\end{aligned}$$

is convergent to zero. Since $T(\cdot)$ is Lipschitz continous, $T(\phi(P_{O_k}(\mathcal{B}(x_k)), P_{\hat{O}_k}(\mathcal{B}(x_k)^H)))$ is converge to $T(\phi(P_{X_k}(\mathcal{B}(x_k)), P_{\hat{X}_k}(\mathcal{B}(x_k)^H)))$. Lemma 2 is proved.

Remark 3: This lemma underpins the convergence of the subspace-approximated SVT-PGD algorithm by establishing a bridge between the approximate and true column subspaces of W_k and W_k^H. The convergence claim hinges on the premise that the approximations provided by O_k and $\hat{O}_k$ sufficiently capture the essence of the true subspaces X_k and $\hat{X}_k$, thereby ensuring that the iteration mechanism effectively moves towards the algorithm's fixed point.

In the following part, the convergence of $T(x_k) = T(\phi(P_{X_k}(\mathcal{B}(x_k)), P_{\hat{X}_k}(\mathcal{B}(x_k)^H)))$ would be proved. Denote the iteration via approximate subspace as $x_{k+1} = T(\phi(P_{O^k}(\mathcal{B}(x_k)), P_{\hat{O}^k}(\mathcal{B}(x_k)).)$. Then, based on Lemma 2, the error between x_{k+1} and $T(x_k)$ is convergent with iteration steps, i.e.,

$$e_k \to 0, \quad k \to \infty. \qquad (57)$$

Step (a): $T(x_k) - x_k$ is bounded with $k \to \infty$.

Denote the error between x_k and $T(x_k)$ as $r_k = T(x_k) - x_k$, $\delta_k = x_{k+1} - x_k$, $\beta_k = T(x_{k+1}) - T(x_k)$, then we have

$$\begin{aligned}
||r_{k+1}||_2^2 &= ||r_{k+1} - r_k + r_k||_2^2 \\
&= ||r_k||_2^2 + ||r_{k+1} - r_k||_2^2 + 2<r_k, r_{k+1} - r_k> \\
&= ||r_k||_2^2 + ||\beta_k - \delta_k||_2^2 + 2<\delta_k - e_k, \beta_k - \delta_k>. \qquad (58)
\end{aligned}$$

When T is $\alpha-$averaged, we have

$$\begin{aligned}
&2 < \delta_k, \beta_k - \delta_k >= ||\beta_k||_2^2 - ||\delta_k||_2^2 - ||\beta_k - \delta_k||_2^2 \\
&= ||T(x_{k+1}) - T(x_k)||_2^2 - ||x_{k+1} - x_k||_2^2 - ||\beta_k - \delta_k||_2^2 \\
&\leq -\frac{1-\alpha}{\alpha}||(I-T)x_{k+1} + (I-T)x_k||^2 - ||\beta_k - \delta_k||^2 \\
&\leq -\frac{1-\alpha}{\alpha}||T(x_{k+1}) - T(x_k) - (x_{k+1} - x_k)
\end{aligned}$$

$$\begin{aligned}&-2T(x_{k+1})+2x_{k+1}||^2-||\beta_k-\delta_k||^2\\&\leq-\frac{1-\alpha}{\alpha}||2(x_{k+1}-T(x_{k+1}))||^2-\frac{1}{\alpha}||\beta_k-\delta_k||^2\\&\leq-\frac{1}{\alpha}||\beta_k-\delta_k||^2,\end{aligned}\tag{59}$$

where the first inequality is due to the fact that T is $\alpha-$averaged, the third inequality is due to the triangle inequality. Hence, we have

$$\begin{aligned}||r_{k+1}||_2^2&\leq||r_k||_2^2-\frac{1-\alpha}{\alpha}||\beta_k-\delta_k||_2^2-2<e_k,\beta_k-\delta_k>\\&=||r_k||_2^2-\frac{1-\alpha}{\alpha}||\beta_k-\delta_k+\frac{\alpha}{\alpha-1}e_k||^2+||\frac{\alpha}{\alpha-1}e_k||^2\\&\leq||r_k||_2^2+||\frac{\alpha}{\alpha-1}e_k||^2.\end{aligned}\tag{60}$$

Applying the above inequality recursively, we have

$$||r_{k+1}||_2^2\leq||r_0||_2^2+\frac{\alpha}{\alpha-1}\sum_k||e_k||^2\tag{61}$$

.

Since the summation of the geometric series with ratio $\frac{\gamma_{r+1}}{\gamma_r}$ is bounded, we have

$$||r_k||<\infty,\quad k\to\infty.\tag{62}$$

Furthermore, we need to prove that r_k is convergent to zero with iteration steps.

Step (b): $T(x_k)-x_k$ is convergent with $k\to\infty$.

Let x^* be a fixed point of T. We have

$$\begin{aligned}\|Tx_k-x_*\|&=\|Tx_k-Tx_*\|\leq L_T\|x_k-x_*\|\\&\leq L_T\big(\big\|Tx_{k-1}-x^*\big\|+\|e_{k-1}\|\big).\end{aligned}\tag{63}$$

Applying the above inequality recursively, we have

$$\left\|Tx^k-x^*\right\|\leq L_T\left(\left\|Tx^0-x^*\right\|+\sum_{i=0}^{k-1}\left\|e^k\right\|\right)<\infty,\tag{64}$$

where the last inequality is due to the fact that $\|e^k\|$ is bounded. Hence, $\|Tx^k-x^*\|$ is bounded for any k. According to Lemma 2 and the Cauchy inequality, we have

$$\begin{aligned}\|x_{k+1}-x_*\|^2&=\|Tx_k-x_*\|^2+\|e_k\|^2+2\langle T(x_k)-x_*,e_k\rangle\\&\leq\|x_k-x_*\|^2-\frac{1-\alpha}{\alpha}\|r_k\|^2+\|e_k\|^2+2\|Tx_k-x_*\|\|e_k\|.\end{aligned}\tag{65}$$

By taking the sum from $k=0$ to m, we obtain

$$\begin{aligned}\frac{1-\alpha}{\alpha}\sum_{i=0}^{m}\left\|r^i\right\|^2&\leq\|x_0-x_*\|^2-||x_{m+1}-x_*||^2\\&\quad+\sum_{i=0}^{m}\|e_i\|^2+2\sum_{i=0}^{m}\|Tx_i-x_*\|\|e_i\|\\&\leq\|x_0-x_*\|^2+\sum_{i=0}^{m}\|e_i\|^2+2\sum_{i=0}^{m}\|Tx_i-x_*\|\|e_i\|.\end{aligned}\tag{66}$$

Therefore, it holds that

$$\begin{aligned}\frac{1-\alpha}{\alpha}\sum_{i=0}^{\infty}\|r_i\|^2&\leq\|x_0-x_*\|^2+\sum_{i=0}^{\infty}\|e_i\|^2\\&\quad+2\sum_{i=0}^{\infty}\|T(x_i)-x_*\|\|e_i\|<\infty,\end{aligned}\tag{67}$$

where the last inequality follows from the fact that $\|T(x_i)-x_*\|$ is bounded, which further implies $\|r_k\|\to0$. The proof is completed.

V. SIMULATION RESULTS

A. PERFORMANCE METRICS AND SIMULATION CONFIGURATIONS

This section presents a comparative analysis of the performance of the SA-SVT-PGD algorithm against the ADMM and SOI-SVT-PGD algorithms. The evaluation criteria include NMSE, the lower bound of achievable spectral efficiency (ASE), and computation overhead.

The NMSE assesses the accuracy of channel estimation across multiple samples, which is defined as:

$$\mathrm{NMSE}=\frac{1}{D}\sum_{i=1}^{D}\frac{\|\mathbf{H}_i-\hat{\mathbf{H}}_i\|_F^2}{\|\mathbf{H}_i\|_F^2},\tag{68}$$

where D represents the total number of channel samples, $\mathbf{H}_i$ is the actual channel matrix for the i-th sample, and $\hat{\mathbf{H}}_i$ denotes the corresponding estimated channel matrix. The lower bound of ASE reflects the system's approximated spectral efficiency as the channel estimation error is considered, which is given by [41]

$$\mathrm{ASE}=\frac{1}{D}\sum_{i=1}^{D}\left\{\log_2\det\left(\mathbf{I}_N+\left(L\frac{N_0}{P_t}+\mathrm{NMSE}\right)^{-1}\mathbf{H}_i\mathbf{H}_i^H\right)\right\},\tag{69}$$

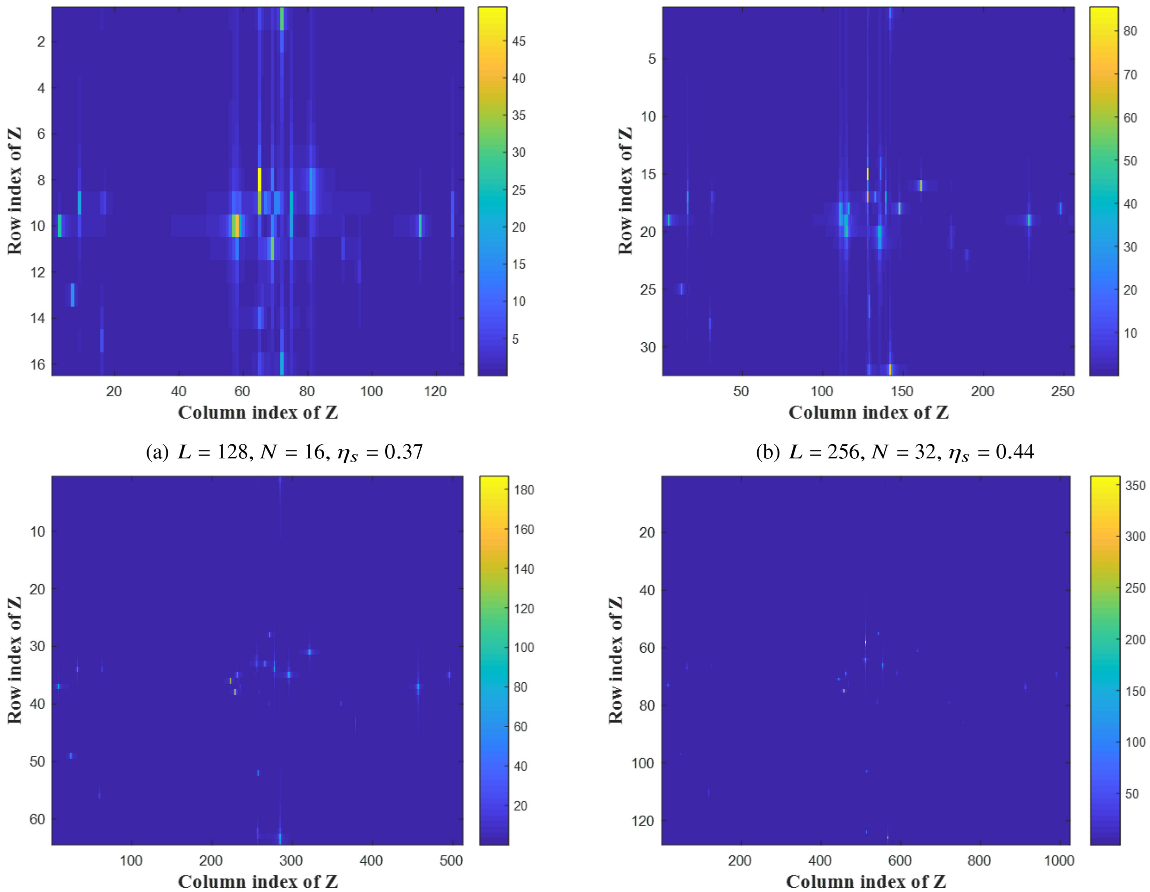
where P_t is the total transmission power, N_0 denotes the power of the additive white Gaussian noise.

In the simulation, the ELAA channels are normalized by configuring $\alpha_l,\forall l$ to meet the complex Gaussian distribution with zero means and unit variance. The azimuth and elevation of the transmitting and receiving antennas follow a Laplace distribution with a mean of 50 degrees and a variance of 90 degrees. 100 channel matrices are sampled to evaluate the NMSE, i.e., $D=100$. λ_1 and λ_2 are configured as 0.5 and 0.01, respectively. $\frac{P_t}{N_0}$ is configured as 15 dB. The stepsize of SA-SVT-PGD and ADMM algorithm, i.e., t and ρ, is configured as 0.1.

B. PERFORMANCE EVALUATIONS

1) QUANTITATIVE CHARACTERIZATIONS FOR SPARSITY OF *Z*

Figure 4a-4d shows the magnitude heat map of Z as the function of the number of antennas. To provide a more intuitive understanding of the sparsity of Z, the proportion of the elements in Z with magnitudes below 0.1, denoted as η_s,



(a) $L = 128$, $N = 16$, $\eta_s = 0.37$

(b) $L = 256$, $N = 32$, $\eta_s = 0.44$

(c) $L = 512$, $N = 64$, $\eta_s = 0.59$

(d) $L = 1024$, $N = 128$, $\eta_s = 0.76$

FIGURE 4. Magnitude of Z as the function of the number of antennas. The number of propagation paths is configured as 25.

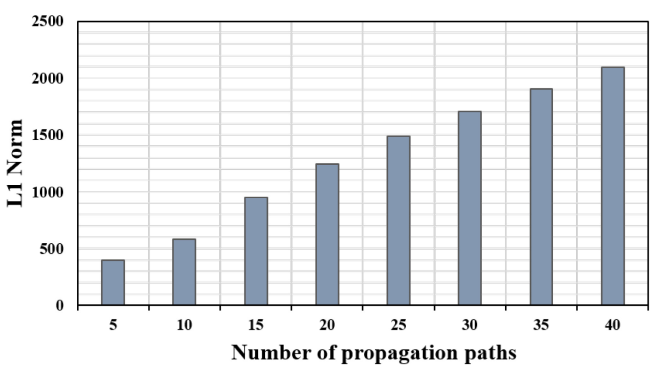


FIGURE 5. L_1 norm of Z as a function of the number of propagation paths.

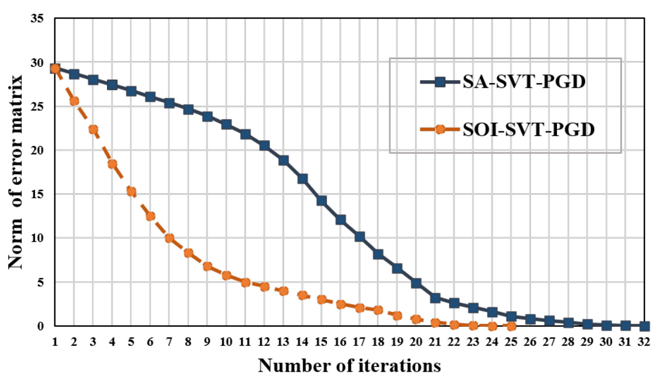


FIGURE 6. ϵ_i of SA-SVT-PGD and SOI-SVT-PGD algorithms versus the number of iterations. $L = 1024$, $N = 128$.

is calculated and shown at each sub-figure. It is observed that the η_s decreases with the increase of the number of ELAA antennas. This result indicates that the sparsity of beam-domain representation of the ELAA channel is enhanced with the increase in the number of antennas.

Figure 5 shows the L_1 norm of Z as a function of the number of propagation paths. It is observed that the L_1 norm of Z increases with the increase of the number of propagation paths. This result indicates that the minimization of the L_1 norm of Z is an effective approach to complete the channel matrix as the number of propagation paths is limited.

2) CONVERGENCE EVALUATIONS

Let $\epsilon_k = ||\mathbf{H}^k - \mathbf{H}^{k+1}||_F$ as the Frobenius norm of error between $\mathbf{H}^k$ and $\mathbf{H}^{k+1}$. In our simulation, the convergence is achieved as $\epsilon_k \leq 0.01$. Figure 6 illustrates the ϵ_i versus the number of iterations for both the SA-SVT-PGD and

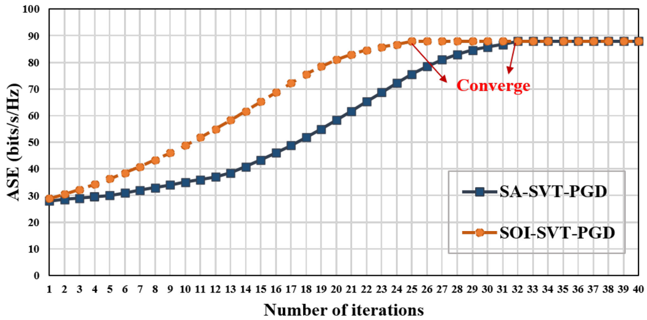


FIGURE 7. ASE as a function of the iterative number for SA-SVT-PGD and SOI-SVT-PGD algorithms.

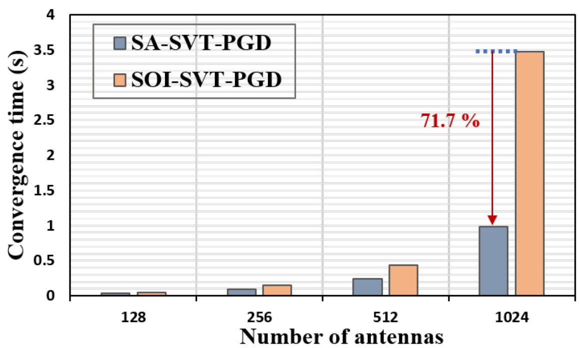


FIGURE 8. Convergence time as a function of the number of antennas for the SA-SVT-PGD and SOI-SVT-PGD algorithms. Missing rate = 0.4, $\mathcal{L} = 25$.

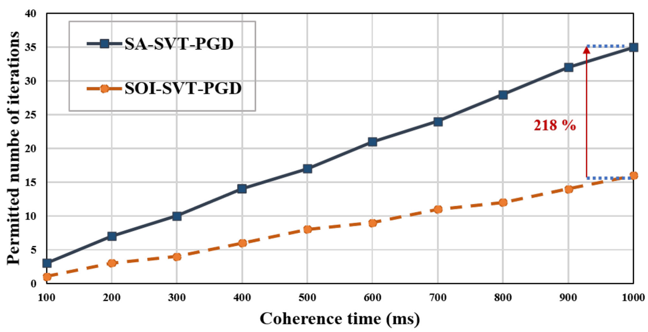


FIGURE 9. Permitted number of iterations as a function of the coherence time for SA-SVT-PGD and SOI-SVT-PGD algorithms.

SOI-SVT-PGD algorithms. The figure clearly demonstrates that both the SA-SVT-PGD and SOI-SVT-PGD algorithms converge with the increase in the number of iterations.

Figure 7 shows the lower bound of ASE as a function of the iterative number for SA-SVT-PGD and SOI-SVT-PGD algorithms. It is observed that the ASE increases with the increase of iterative numbers and is saturated at 25 and 32 iterations for the SOI-SVT-PGD and SA-SVT-PGD algorithms, respectively.

Figure 8 illustrates the convergence time as a function of the number of antennas for the SA-SVT-PGD and SOI-SVT-PGD algorithms. The results indicate that the convergence time increases with the number of antennas, which aligns with the intuitive observation that a larger channel matrix

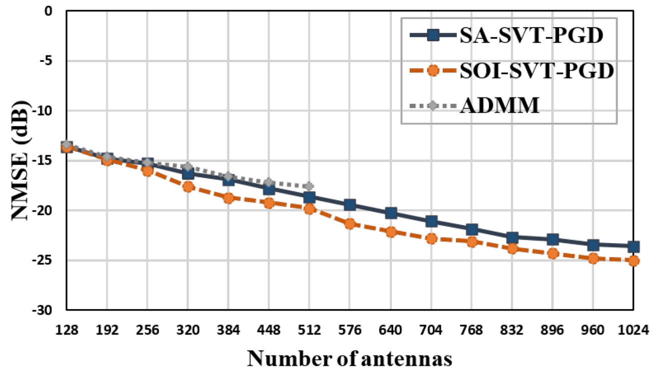


FIGURE 10. NMSE performance as a function of the number of antennas for the SA-SVT-PGD, SOI-SVT-PGD, and ADMM algorithms. $\mathcal{L} = 25$, missing rate = 0.4.

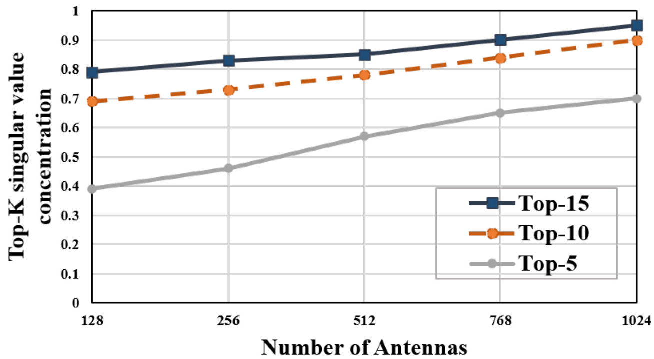


FIGURE 11. Top-K(H) singular value concentration versus the number of antennas L at $K = 5, 10, 15$. $\mathcal{L} = 25$, $N = \frac{L}{8}$.

size results in higher computational overhead. Notably, for a typical configuration of 1024 antennas in practical ELAA communication systems, the convergence time of the SA-SVT-PGD algorithm is reduced by up to 71.7% compared to the SOI-SVT-PGD algorithm.

Figure 9 shows the permitted number of iterations as a function of the coherence time for SA-SVT-PGD and SOI-SVT-PGD algorithms. When the coherence time is configured at 1000 ms, the permitted iterative number of the SA-SVT-PGD algorithm is 218% of that of the SOI-SVT-PGD algorithm.

3) NMSE AND ASE EVALUATIONS

Figure 10 illustrates the NMSE performance as a function of the number of antennas for the SA-SVT-PGD, SOI-SVT-PGD, and ADMM algorithms. The results indicate that the NMSE decreases with the increase in the number of antennas. Specifically, when the number of antennas reaches 1024, the SOI-SVT-PGD and SA-SVT-PGD algorithms achieve NMSE values of -25 dB and -23.6 dB, respectively. In contrast, the ADMM algorithm exhibits scalability limitations when the number of antennas exceeds 512, primarily due to the memory requirements surpassing the available memory capacity in ELAA communication systems.

To better illustrate the reason why NMSE decreases with the increase in the number of antennas, Figure 11 shows

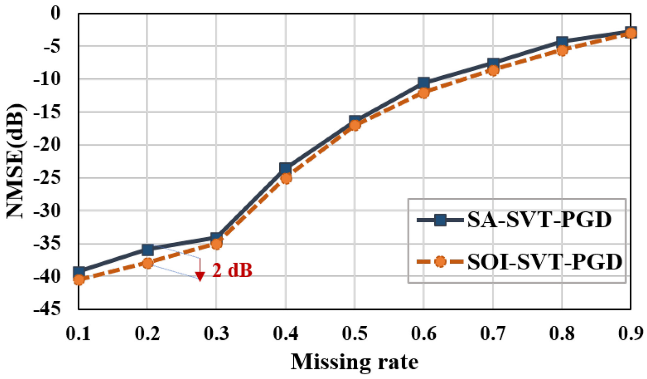


FIGURE 12. NMSE performance as a function of the missing rate for the SA-SVT-PGD and SOI-SVT-PGD algorithms. $\mathcal{L} = 25$, $L = 1024$.

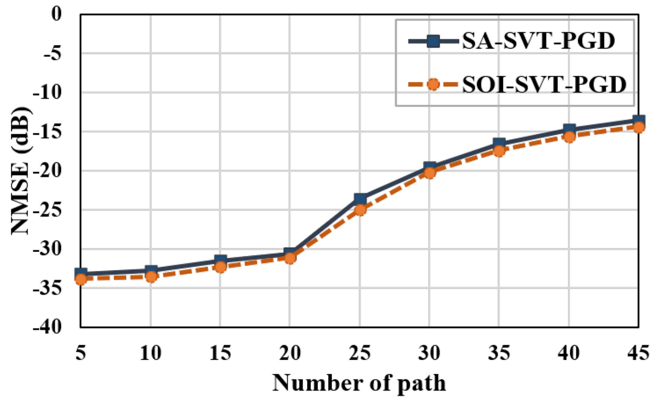


FIGURE 13. NMSE performance as a function of the number of path for the SA-SVT-PGD and SOI-SVT-PGD algorithms. Missing rate = 0.4, $L = 1024$.

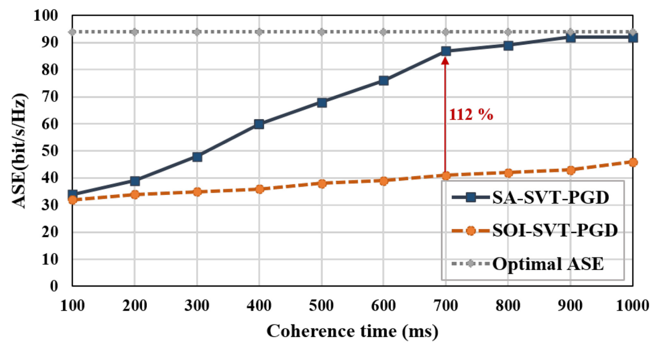


FIGURE 14. ASE as a function of the given coherence time for the SA-SVT-PGD and SOI-SVT-PGD algorithms. Missing rate = 0.4, $L = 1024$, $\mathcal{L} = 25$.

Top-$K(\mathbf{H})$ singular value concentration versus the number of antennas L at $K = 5, 10, 15$. The Top-K singular values concentration is represented by [42]

$$\text{Top-}K(\mathbf{H}) = \frac{\sum_{i\in\mathcal{K}} \sigma_i}{\sum_{j\in\mathcal{R}} \sigma_j}, \tag{70}$$

where K is the number of dominant singular values considered in the summation, $\mathcal{K}$ represents the set of indices corresponding to the Top-K singular values, and $\mathcal{R}$ is the full set of indices corresponding to all singular values of $\mathbf{H}$. Extensive research shows that the NMSE of the low-rank matrix completion algorithm decreases with the increases of Top-$K(\mathbf{H})$ [42], [43]. It is observed that the Top-5, Top-10, and Top-15 singular value concentrations increase with the increase in the number of antennas.

Figure 12 illustrates the NMSE performance as a function of the missing rate for the SA-SVT-PGD and SOI-SVT-PGD algorithms. The results demonstrate that the NMSE increases with the missing rate, which aligns with the intuitive observation that a higher availability of elements leads to a more accurate channel matrix estimation. Notably, the maximum performance gap between the two algorithms is limited to 2 dB when the missing rate is configured to 0.2, which is attributed to the convergence of the approximated subspace of SA-SVT-PGD to the ground-truth subspace of the channel's Gram matrix.

Figure 13 illustrates the NMSE performance as a function of the number of path for the SA-SVT-PGD and SOI-SVT-PGD algorithms. The results demonstrate that the NMSE increases with the increase in the number of paths, which is attributed to the corresponding increase in the rank of the ELAA channel.

Figure 14 illustrates the ASE as a function of the given coherence time for the SA-SVT-PGD and SOI-SVT-PGD algorithms, with the optimal ASE without channel estimation error serving as the upper bound. The results show that the ASE increases with the coherence time, which aligns with the intuitive observation that a longer coherence time allows for more iterations of the SA-SVT-PGD and SOI-SVT-PGD algorithms. Specifically, when the coherence time is set to 700 ms, the SA-SVT-PGD algorithm improves the ASE by 112% compared to the SOI-SVT-PGD algorithm.

VI. CONCLUSION

This study introduces a novel low-complexity matrix completion approach tailored for complete channel estimation within ELAA communication systems. By addressing both the sparsity and low-rank characteristics of the ELAA channel matrix, we have formulated an optimization problem that jointly minimizes the nuclear and L_1 norms. To scale the matrix completion algorithm to ELAA communication systems, the SVT-PGD algorithm is designed by alternatively applying the SVT and soft-thresholding operators to the observed channel matrix. To further reduce the computational overhead of SVT in the SVT-PGD algorithm, an SA-SVT-PGD algorithm is developed to exploit the column space similarity of the channel matrix in the consecutive PGD iterations. Theoretical analysis has validated the convergence of the proposed SA-SVT-PGD, ensuring its robustness and reliability. Through rigorous simulation studies, we demonstrated that the SA-SVT-PGD could reduce the convergence time by a maximum of 71.7% and improve the ASE by a maximum of 112% in the given coherence time compared to the SOI-SVT-PGD algorithm.

Looking ahead, we aim to broaden the applicability of our algorithm by integrating it into the joint beamforming and channel estimation tasks. Such advancements promise to alleviate the limitations of CSI-sharing traffic in forthcoming generations of ELAA systems. Furthermore, we envision extending our approach to other mobile network architectures, including radio stripes and cloud-radio access network (C-RAN) structures, thereby contributing to the evolution of efficient, high-capacity wireless communication infrastructures.

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